

Advanced Banana Ripeness, Quality Detection, and Counting Using Custom CNN, Transfer Learning, YOLO Object Detection, and Streamlit

Project Report

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Certification: Data Science with Artificial Intelligence

Executive Summary

This project was completed as the capstone project for the Data Science with Artificial Intelligence Certification Program. The objective was to develop a complete computer vision solution capable of classifying banana ripeness, detecting bananas within an image, and automatically counting them using deep learning techniques.

The project required implementing multiple deep learning approaches rather than relying on a single model. Accordingly, three image classification models—a Custom Convolutional Neural Network (CNN), MobileNetV2, and EfficientNetB0—were developed and compared. In addition, a YOLOv8 object detection model was trained to identify bananas within an image and estimate the total inventory count. Finally, all trained models were integrated into a Streamlit web application that supports image upload and live camera capture for real-time inference.

Throughout the project, I gained practical experience in image preprocessing, dataset preparation, model training, transfer learning, object detection, model evaluation, and deployment. The comparative study also highlighted an important machine learning concept: a model's confidence score does not always indicate that its prediction is correct. During external testing, MobileNetV2 correctly classified a sample image, while the Custom CNN and EfficientNetB0 produced incorrect predictions despite reporting confidence scores.

Overall, this project successfully achieved its learning objectives by providing hands-on experience with the complete deep learning development lifecycle and demonstrating both the strengths and limitations of computer vision models in real-world applications.

1. Introduction

Manual quality assessment of agricultural products is an important process throughout food production, warehousing, retail distribution, and supply chain management. Bananas are typically graded based on visual characteristics such as peel colour, surface texture, bruising, and visible deterioration. Since these inspections are performed manually, they are often subjective, time-consuming, and prone to inconsistencies.

Recent developments in artificial intelligence and computer vision have enabled many of these repetitive inspection tasks to be automated using deep learning models. Convolutional Neural Networks (CNNs) have demonstrated strong performance in image classification tasks, while transfer learning models can leverage pretrained knowledge to improve accuracy even with relatively small datasets. Similarly, object detection models such as YOLO provide the ability to locate multiple objects within an image while simultaneously estimating their quantity.

The primary objective of this project was to implement an end-to-end deep learning solution capable of performing banana ripeness classification and banana detection. More importantly, as part of my certification program, the project provided an opportunity to understand every stage of a real-world machine learning workflow—from dataset preparation and model training to deployment and evaluation.

To make the implementation evidence clear for certification assessment, the report includes actual project outputs such as:

- dataset sample images,
- training and validation accuracy/loss plots,
- confusion matrices,
- YOLOv8 detection results,
- Streamlit application screenshots,
- and comparison test outputs from unseen images.

These visual artifacts provide proof of implementation and help demonstrate that the project was not only designed conceptually but also executed as a working deep learning application.

2. Project Objectives

The project objectives were derived from the certification requirements and included:

- Develop a Custom CNN model for banana ripeness classification.
- Implement transfer learning models using MobileNetV2 and EfficientNetB0.
- Compare the performance of all classification models.
- Train a YOLOv8 model for banana detection and automated counting.
- Develop a Streamlit web application supporting image upload and live camera capture.
- Evaluate model performance using real-world test images.
- Gain practical experience in building and deploying computer vision applications.
- Present implementation evidence through training outputs, screenshots, and comparison results for certification review.

3. Dataset and Methodology

The project used a labelled banana image dataset consisting of four ripeness categories:

- Unripe
- Ripe
- Overripe
- Rotten

Before training, the dataset was organized into training, validation, and testing folders. Images were resized to the required input dimensions for each model, normalized to improve learning efficiency, and augmented using rotation, zooming, flipping, and brightness adjustments. These preprocessing techniques increased dataset diversity and improved model generalization.

3.1 Dataset Overview

The dataset contained banana images captured under different lighting conditions, orientations, and backgrounds. This variation helped the models learn more robust visual features rather than memorizing a single image style.

Insert the following actual dataset evidence in the report:

These figures help demonstrate that the dataset was properly organized before model training.

3.2 Data Preprocessing and Augmentation

Preprocessing was an important step because deep learning models require consistent input dimensions and normalized pixel values. The images were resized according to the model architecture requirements and scaled to improve convergence during training.

Data augmentation was applied to reduce overfitting and improve generalization. The augmentation pipeline included:

- horizontal flipping,
- small rotations,
- zooming,
- brightness variation,
- and slight shifts in image position.

Insert actual augmentation examples here:

This figure should show how the same banana image appears after augmentation, demonstrating the diversity introduced during training.

3.3 Model Development Approach

Three different image classification approaches were implemented.

The first model was a Custom CNN developed from scratch using TensorFlow and Keras. This model established the baseline for comparison and provided valuable insight into how convolutional layers extract image features directly from training data.

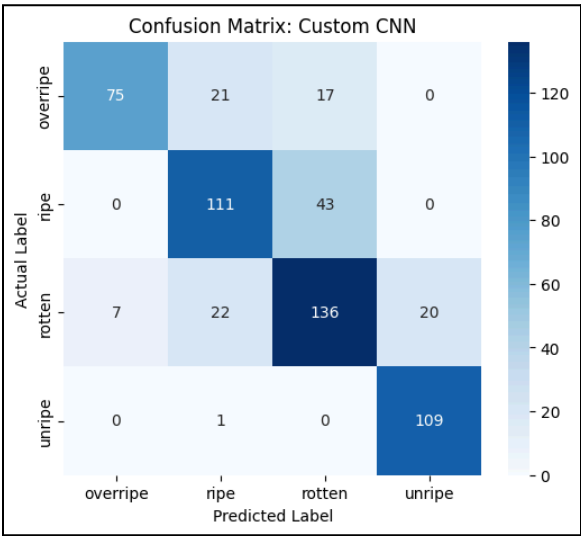
The second and third models used transfer learning. MobileNetV2 was selected because of its lightweight architecture and suitability for real-time inference, while EfficientNetB0 was chosen for its improved feature extraction capabilities through compound scaling. Both models were fine-tuned on the banana dataset after initializing with pretrained ImageNet weights.

In addition to image classification, YOLOv8 was trained to detect bananas within an image and automatically count the total number of visible fruits. This extended the project beyond classification by introducing object localization and inventory estimation.

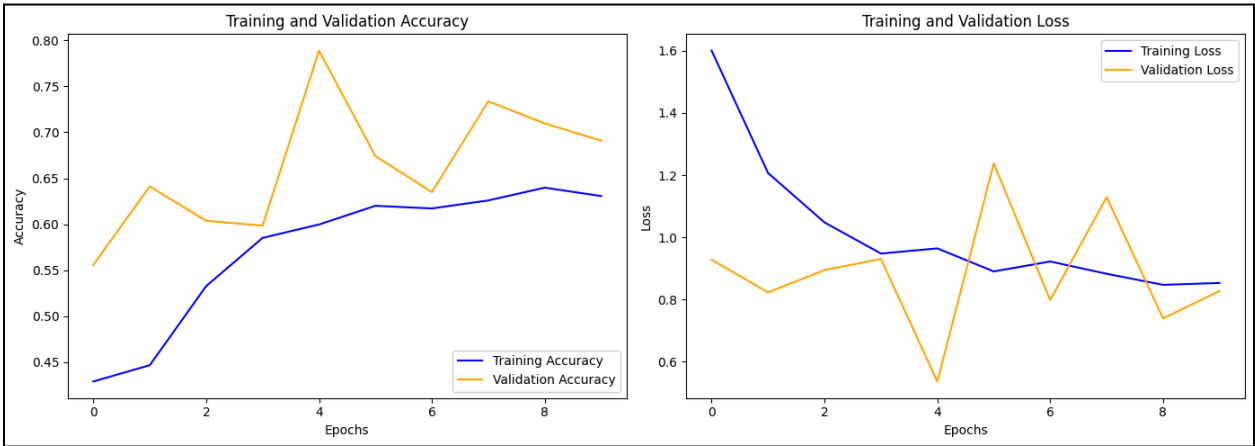
Finally, all trained models were integrated into a Streamlit web application that enables users to upload images or capture live photographs for immediate prediction.

3.4 Training Evidence for Classification Models

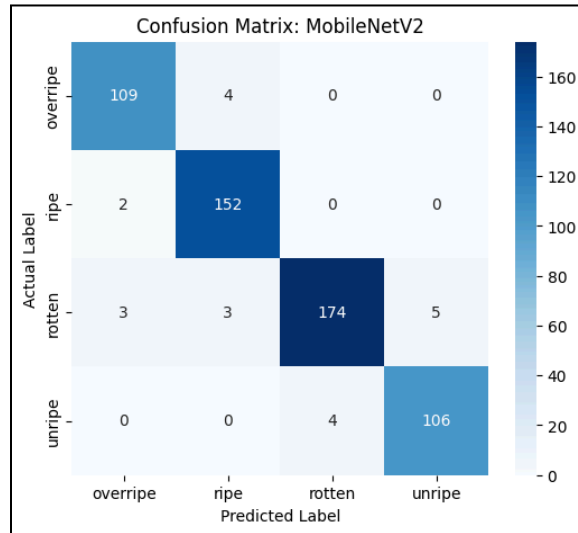
Custom CNN Training Output



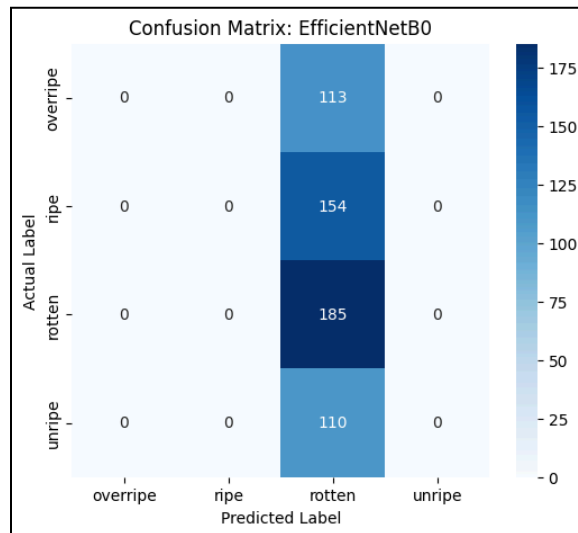
Training and Validation: Accuracy, Loss



MobileNetV2 Training Output



EfficientNetB0 Training Output



These figures provide direct evidence of model training behaviour.

3.5 YOLOv8 Training Evidence

The YOLOv8 model was trained separately for banana detection and counting. The following outputs should be included:



3.6 Streamlit Application Implementation

The trained models were deployed in a Streamlit application to provide a user-friendly interface for inference. The application supports:

- image upload,
- live camera capture,
- model selection,
- prediction display,
- confidence score display,
- and banana counting through YOLO detection.

Insert actual implementation screenshots here:

These screenshots are important because they show the final working application rather than only the training process.

4. Results and Discussion

The project demonstrated that different deep learning architectures behave differently when presented with the same unseen image.

During one of the real-world tests, the uploaded image contained ripe bananas. The three classification models produced different outputs:

Custom CNN Model Output

Spaces

dcvhere / banana-ripeness-classifier

Like

Running

Logs

AppFilesCommunitySettings

Control Panel

Select Classification Model:
Custom CNN

Select Image Input Method:
File Upload
Camera Capture

Upload an image (JPG/PNG)
images.jpeg
5.0KB

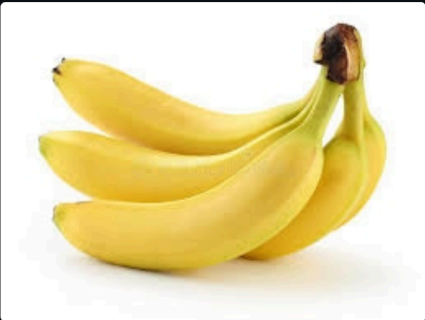


Advanced Banana Ripeness & Quality Detection

Welcome to the automated banana grading system. **User Guidance:** For best results, use clear lighting and ensure the bananas are fully visible in the frame.

Input Image

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.



Analysis Results

Selected Model: Custom CNN

Predicted Quality: Rotten

Confidence Score: 92.87%

Detection View

Total Bananas Detected
0

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.



MobileNetV2 Model Output

Spaces

dcvhere / banana-ripeness-classifier

Like

Running

Logs

AppFilesCommunitySettings

Control Panel

Select Classification Model:
MobileNetV2

Select Image Input Method:
File Upload
Camera Capture

Upload an image (JPG/PNG)
images.jpeg
5.0KB



Advanced Banana Ripeness & Quality Detection

Welcome to the automated banana grading system. **User Guidance:** For best results, use clear lighting and ensure the bananas are fully visible in the frame.

Input Image

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.



Analysis Results

Selected Model: MobileNetV2

Predicted Quality: Ripe

Confidence Score: 96.76%

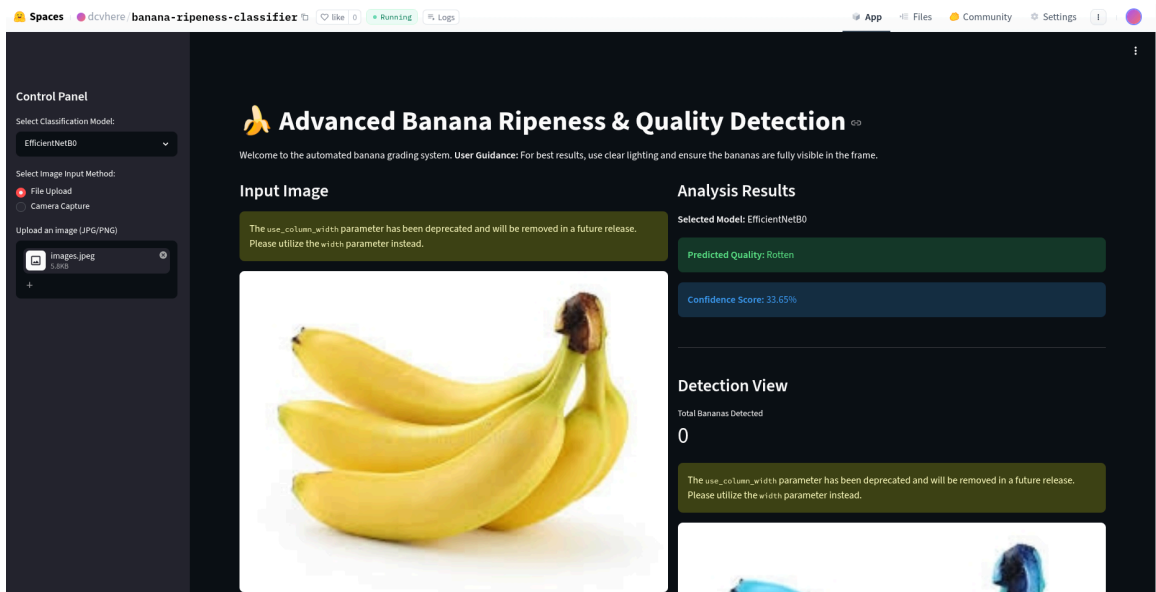
Detection View

Total Bananas Detected
0

The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the width parameter instead.



EfficientNetB0 Model Output



Output Comparison Table

Model	Prediction	Confidence
Custom CNN	Rotten	92.87%
MobileNetV2	Ripe	96.76%
EfficientNetB0	Rotten	33.65%

Among the evaluated models, **MobileNetV2** correctly classified the sample image as **Ripe** with the highest confidence score. The Custom CNN incorrectly classified the image as **Rotten** despite assigning a confidence score above 92%, while EfficientNetB0 also predicted **Rotten**, although with a significantly lower confidence score that reflected greater uncertainty.

To strengthen this section, the actual comparison test outputs should be inserted as screenshots from the application or notebook:

This experiment demonstrated one of the most important lessons learned during the project: **confidence scores represent the model’s certainty in its prediction, not the correctness of the prediction itself**. Therefore, model evaluation should not rely solely on confidence values but should also consider validation accuracy, confusion matrices, and testing on previously unseen images.

The YOLOv8 object detection model successfully detected bananas and generated bounding boxes for inventory counting. Although its performance depended on

image quality and object visibility, the detector demonstrated the practicality of combining image classification with object detection within a single application.

Insert actual YOLO detection comparison outputs here:

Overall, the project showed that transfer learning generally produced better generalization than the Custom CNN baseline. MobileNetV2 emerged as the most reliable classification model during external testing, while YOLOv8 successfully extended the application with automated object detection and counting capabilities.

5. Learning Outcomes and Challenges

This project provided valuable practical experience across the complete machine learning lifecycle.

From a technical perspective, I learned how to prepare image datasets, implement convolutional neural networks, apply transfer learning, train YOLO object detection models, evaluate classification performance, and deploy deep learning models using Streamlit and Hugging Face Spaces.

One of the most valuable lessons was understanding that high validation accuracy alone does not guarantee reliable real-world performance. During testing, different models produced different predictions for the same image, highlighting the importance of evaluating models using multiple datasets rather than relying solely on training metrics.

The project also improved my understanding of image preprocessing, hyperparameter tuning, overfitting, model generalization, and deployment challenges. Integrating multiple trained models into a single application required maintaining consistent preprocessing pipelines and efficient model loading procedures.

In addition to technical skills, the project strengthened my ability to troubleshoot implementation issues, interpret model behaviour, and critically evaluate experimental results instead of assuming that higher confidence always implies better predictions.

To support certification evaluation, the following evidence should also be included in the appendix or final pages:

- training logs,
- notebook screenshots,
- model summary outputs,
- saved model files,
- and deployment screenshots from Hugging Face or local Streamlit execution.

6. Limitations and Future Scope

Although the project achieved its primary objectives, several limitations remain.

The dataset mainly consisted of standard Cavendish bananas, which may reduce classification accuracy when different banana varieties are presented. Lighting conditions, camera quality, image backgrounds, and partial occlusions can also influence prediction accuracy.

The classification models assign discrete ripeness categories even though fruit ripening is a continuous biological process. Consequently, bananas positioned near category boundaries may receive uncertain predictions.

Another limitation is that the project was evaluated on a limited set of test images. While the results were useful for demonstrating model behaviour, a larger and more diverse test set would provide stronger evidence of generalization.

Future improvements may include expanding the dataset to multiple fruit varieties, integrating disease detection, predicting remaining shelf life instead of discrete classes, optimizing the models for edge devices, and integrating cloud-based inventory management systems for commercial deployment.

Additional future enhancements could include:

- adding Grad-CAM or saliency maps for explainability,
- improving YOLO annotation quality for better detection,
- benchmarking inference speed across devices,
- and deploying the application as a production-ready web service.

7. Conclusion

This project successfully fulfilled the requirements of the Data Science with Artificial Intelligence certification by demonstrating the complete development of an end-to-end computer vision application for banana ripeness classification, quality detection, and inventory counting.

The implementation included a Custom CNN, MobileNetV2, EfficientNetB0, YOLOv8 object detection, and a Streamlit-based deployment interface. Rather than focusing solely on achieving the highest possible accuracy, the project emphasized understanding how different deep learning models behave throughout training, validation, deployment, and real-world testing.

The comparative evaluation showed that MobileNetV2 provided the most reliable predictions during external testing, while the Custom CNN and EfficientNetB0 illustrated that high confidence scores do not necessarily correspond to correct predictions. This reinforced the importance of comprehensive model evaluation before deploying machine learning systems in practical applications.

Overall, this project significantly strengthened my understanding of deep learning, computer vision, transfer learning, object detection, and deployment. More importantly, it provided practical experience in developing, evaluating, and deploying AI solutions while encouraging a critical and evidence-based approach to interpreting model performance.